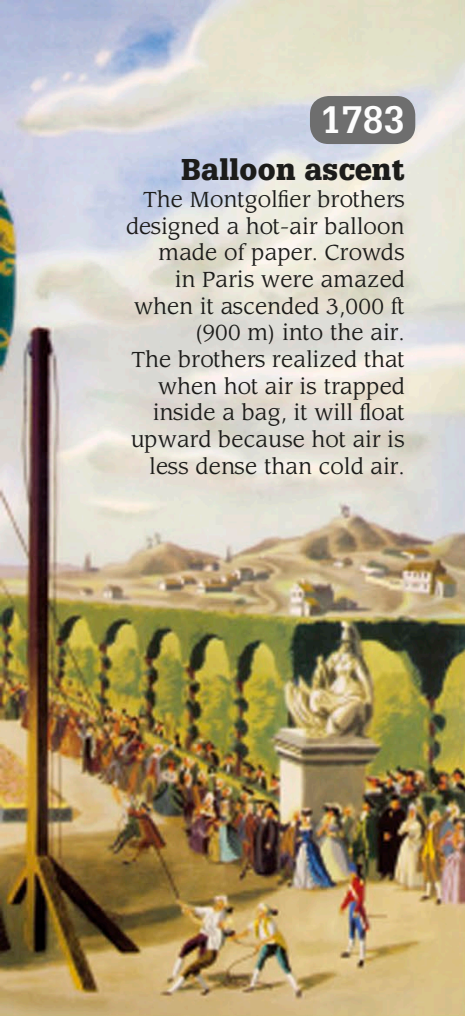


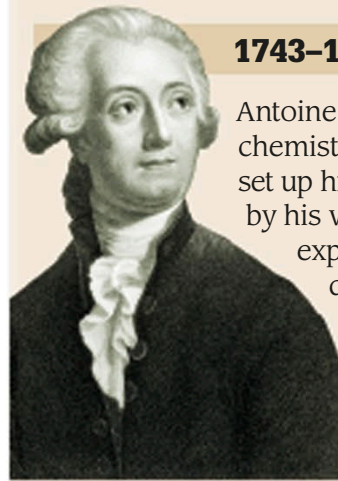
1783

Balloon ascent

The Montgolfier brothers designed a hot-air balloon made of paper. Crowds in Paris were amazed when it ascended 3,000 ft (900 m) into the air. The brothers realized that when hot air is trapped inside a bag, it will float upward because hot air is less dense than cold air.



1743–1794

ANTOINE LAVOISIER

Antoine Lavoisier is regarded as the father of modern chemistry. Born into a wealthy family in Paris, France, he set up his own laboratory to carry out experiments, helped by his wife and several assistants. Known for his precise experiments and careful measurements, Lavoisier discovered the role oxygen played in rusting, combustion (burning), and respiration (breathing).

Experimental scientist

As well as his work in understanding oxygen, Lavoisier also showed that matter is neither created nor destroyed during chemical changes. He was executed during the French Revolution because he had previously worked as a tax inspector for the king.

Elements of Chemistry

Lavoisier published the *Elements of Chemistry*, a book that laid the foundations of chemistry as a science. It contained a list of 33 elements (substances that could not be broken down any further) and introduced modern chemical names, including oxygen and hydrogen.

The *Elements of Chemistry* was published in 1789, the year of the French Revolution.



1785

1781

Seventh planet

William Herschel, a German-born astronomer living in England, identified a new planet with a telescope he had built in his backyard. It was given the name Uranus (the Latin form of Ouranos, the Greek god of the sky). Eight years later, a newly discovered element was called uranium after the planet.



Uranus lies at an angle of more than 90° to the rest of the solar system.

1785

Changing landscapes

Scottish geologist James Hutton wrote that the landscape is continually being shaped by slow-moving natural processes, such as erosion. According to this view, Earth must be millions of years old. His ideas proved correct, though opposed at the time by Christians, who believed that Earth was only 6,000 years old.

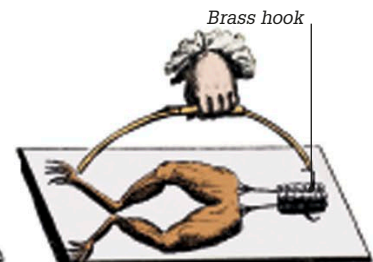
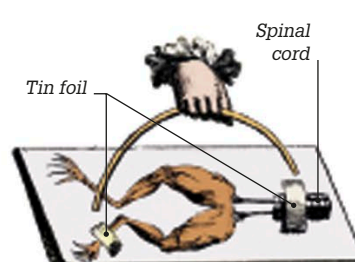
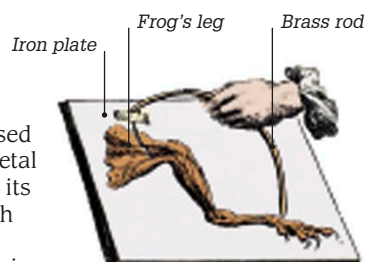
“... little causes... are considered as bringing about the greatest changes of the Earth.”

James Hutton, 1795

1781

Leap frog

When Italian scientist Luigi Galvani connected the exposed nerves of a dead frog to a metal wire during a thunderstorm, its legs twitched with every flash of lightning. His macabre experiments were a key step in understanding electricity, and inspired the novel *Frankenstein*.



Galvani experimented on frogs' legs by lying them on metal plates and touching them with different metals.